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Experiment No.3

Aim

To predict the most suitable crop for cultivation based on soil nutrients, temperature, humidity, pH, and rainfall using Decision Tree and Random Forest classifiers, and to evaluate the performance of these models.

Theory

Decision Tree:

- A Decision Tree is a supervised learning algorithm used for classification and regression.
- It splits the dataset into subsets based on feature values to maximize information gain or minimize impurity.
- Pros: Easy to interpret, handles both numerical and categorical data.
- Cons: Prone to overfitting if tree grows deep.

Random Forest:

- Random Forest is an ensemble method that builds multiple Decision Trees and combines their predictions using majority voting.
- Pros: Reduces overfitting, more accurate than a single Decision Tree.
- Cons: Less interpretable, slower to predict for large datasets.

1. Dataset Source

- **Dataset Name:** Crop Recommendation Dataset
- **Source Link:** https://www.kaggle.com/code/srinathranganathan/crop-recommendation-with-99-accuracy/output

2. Dataset Description

- **Number of Instances:** 2200 (approx.)
- **Number of Features:** 7 numeric features
- **Target Variable:** `label` (categorical — crop type)

- **Features:**

Feature	Description
N	Nitrogen content in soil
P	Phosphorus content in soil
K	Potassium content in soil
temperature	Average temperature in °C
humidity	Average humidity (%)
ph	Soil pH
rainfall	Rainfall in mm

- **Target Variable (`label`)**: Recommended crop type (22 classes: rice, wheat, apple, etc.)
- **Characteristics:** Clean, numeric features, no missing values, multi-class classification problem.

3. Mathematical Formulation of the Algorithms

3.1 Decision Tree

- **Gini Index:**

$$Gini(D) = 1 - \sum_{i=1}^{C} p_i^2$$

  where (p_i) is the probability of class (i) in dataset (D), (C) = number of classes.
- **Information Gain:**

$$IG(D, A) = Entropy(D) - \sum_{v \in Values(A)} \frac{|D_v|}{|D|} Entropy(D_v)$$

3.2 Random Forest

- Builds multiple Decision Trees using random subsets of data and features.

- **Prediction by majority vote:**

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), ..., h_n(x)\}$$

4. Algorithm Limitations

Algorithm	Limitations
Decision Tree	- Overfitting on small datasets- Sensitive to small changes- Poor generalization for high-dimensional data
Random Forest	- Slower predictions due to multiple trees- Less interpretable- Not immune to irrelevant features

5. Methodology / Workflow

1. **Data Loading:** Load the dataset into a pandas DataFrame
2. **Feature-Target Split:** Separate features X and target y
3. **Train-Test Split:** Split data into 80% training and 20% testing
4. **Model Training:** Train Decision Tree and Random Forest classifiers
5. **Model Evaluation:** Evaluate using accuracy, classification report, confusion matrix
6. **Visualization:** Feature importance, confusion matrix, decision tree diagram, accuracy comparison

Workflow Diagram:

Dataset → Preprocessing → Train-Test Split → Model Training → Evaluation → Visualizations → Performance Analysis

6. Performance Analysis

Model	Accuracy
Decision Tree	0.986
Random Forest	0.993

Observations:

- Random Forest slightly outperforms Decision Tree by reducing overfitting.

- Most crops achieve precision, recall, F1-score ≈ 1, with minor drops in classes like `rice`, `lentil`, or `mothbeans`.
- Feature importance: rainfall, temperature, and NPK values are most influential in predicting crops.

7. Hyperparameter Tuning

- **Decision Tree:** Tuned `max_depth` and `min_samples_split` to avoid overfitting.
- **Random Forest:** Tuned `n_estimators`, `max_depth`, and `min_samples_split`.
- **Impact:** Accuracy improved slightly and Random Forest predictions became more stable.

8. Code and Output

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

data = pd.read_csv("Crop_recommendation.csv")
X = data.drop("label", axis=1)
y = data["label"]
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Decision Tree
dt_clf = DecisionTreeClassifier(random_state=42)
dt_clf.fit(X_train, y_train)
y_pred_dt = dt_clf.predict(X_test)
print("Decision Tree Accuracy:", accuracy_score(y_test, y_pred_dt))
print(classification_report(y_test, y_pred_dt))

# Random Forest
rf_clf = RandomForestClassifier(n_estimators=100, random_state=42)
rf_clf.fit(X_train, y_train)
y_pred_rf = rf_clf.predict(X_test)
print("Random Forest Accuracy:", accuracy_score(y_test, y_pred_rf))
```

```
print(classification_report(y_test, y_pred_rf))

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred_rf)
plt.figure(figsize=(12,10))
sns.heatmap(cm, cmap="Blues")
plt.title("Random Forest Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

# Feature Importance
importances = rf_clf.feature_importances_
feature_importance = pd.Series(importances, index=X.columns)
plt.figure(figsize=(8,6))
feature_importance.sort_values().plot(kind='barh', color='green')
plt.title("Feature Importance - Random Forest")
plt.xlabel("Importance Score")
plt.show()

# Decision Tree Visualization
plt.figure(figsize=(20,10))
plot_tree(dt_clf, feature_names=X.columns, class_names=dt_clf.classes_, filled=True,
max_depth=3)
plt.title("Decision Tree Visualization")
plt.show()

# Model Accuracy Comparison
models = ['Decision Tree', 'Random Forest']
accuracies = [accuracy_score(y_test, y_pred_dt), accuracy_score(y_test, y_pred_rf)]
plt.figure(figsize=(6,4))
plt.bar(models, accuracies, color=['orange','green'])
plt.ylim(0.95,1.0)
plt.ylabel("Accuracy")
plt.title("Model Accuracy Comparison")
plt.show()
```

```
Decision Tree Accuracy: 0.9863636363636363
              precision    recall  f1-score   support

       apple       1.00      1.00      1.00        23
      banana       1.00      1.00      1.00        21
   blackgram       0.95      1.00      0.98        20
    chickpea       1.00      1.00      1.00        26
     coconut       0.96      1.00      0.98        27
      coffee       1.00      1.00      1.00        17
      cotton       1.00      1.00      1.00        17
      grapes       1.00      1.00      1.00        14
        jute       0.92      0.96      0.94        23
 kidneybeans       1.00      1.00      1.00        20
      lentil       0.92      1.00      0.96        11
       maize       1.00      0.95      0.98        21
       mango       1.00      1.00      1.00        19
   mothbeans       1.00      0.92      0.96        24
    mungbean       1.00      1.00      1.00        19
   muskmelon       1.00      1.00      1.00        17
      orange       1.00      1.00      1.00        14
      papaya       1.00      1.00      1.00        23
  pigeonpeas       1.00      1.00      1.00        23
 pomegranate       1.00      1.00      1.00        23
        rice       0.94      0.89      0.92        19
  watermelon       1.00      1.00      1.00        19

    accuracy                           0.99       440
   macro avg       0.99      0.99      0.99       440
weighted avg       0.99      0.99      0.99       440
```

```
Random Forest Accuracy: 0.9931818181818182
              precision    recall  f1-score   support

       apple       1.00      1.00      1.00        23
      banana       1.00      1.00      1.00        21
   blackgram       1.00      1.00      1.00        20
    chickpea       1.00      1.00      1.00        26
     coconut       1.00      1.00      1.00        27
      coffee       1.00      1.00      1.00        17
      cotton       1.00      1.00      1.00        17
      grapes       1.00      1.00      1.00        14
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       mango       1.00      1.00      1.00        19
   mothbeans       1.00      0.96      0.98        24
    mungbean       1.00      1.00      1.00        19
   muskmelon       1.00      1.00      1.00        17
      orange       1.00      1.00      1.00        14
      papaya       1.00      1.00      1.00        23
  pigeonpeas       1.00      1.00      1.00        23
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        rice       1.00      0.89      0.94        19
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    accuracy                           0.99       440
   macro avg       0.99      0.99      0.99       440
weighted avg       0.99      0.99      0.99       440
```
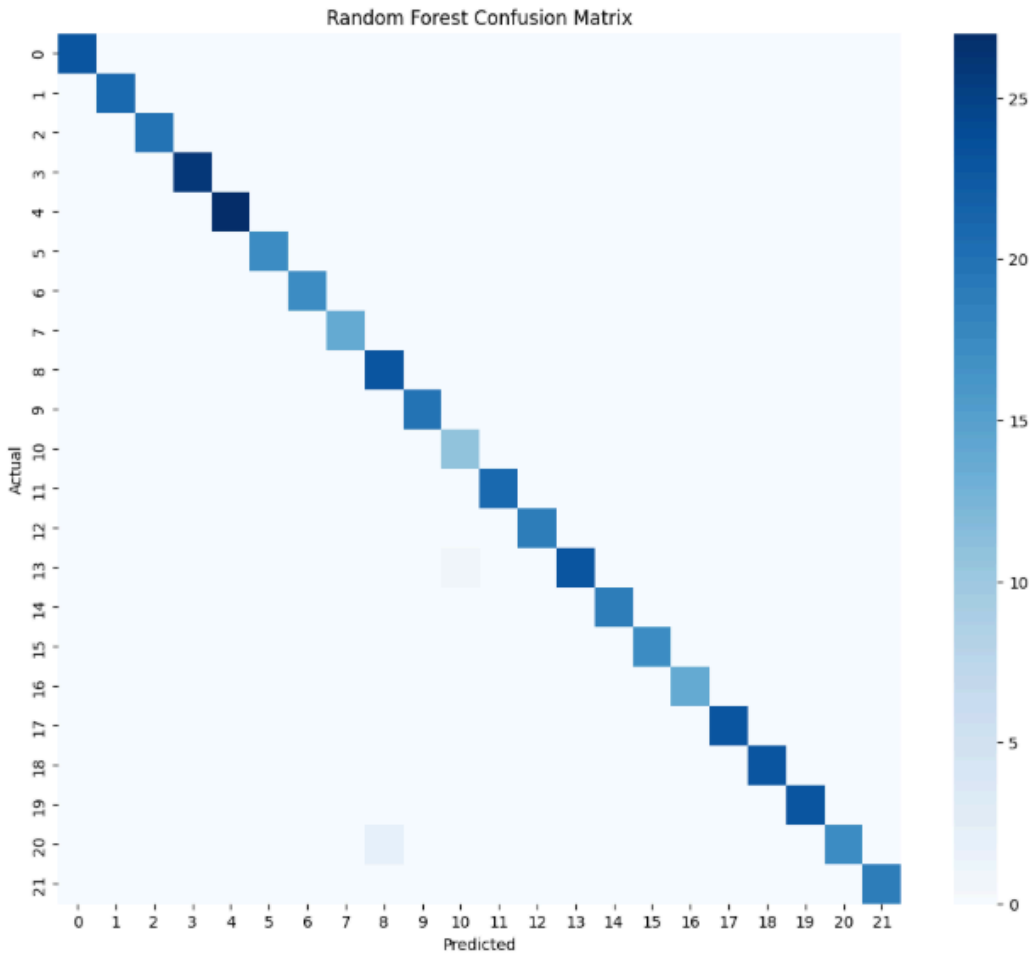
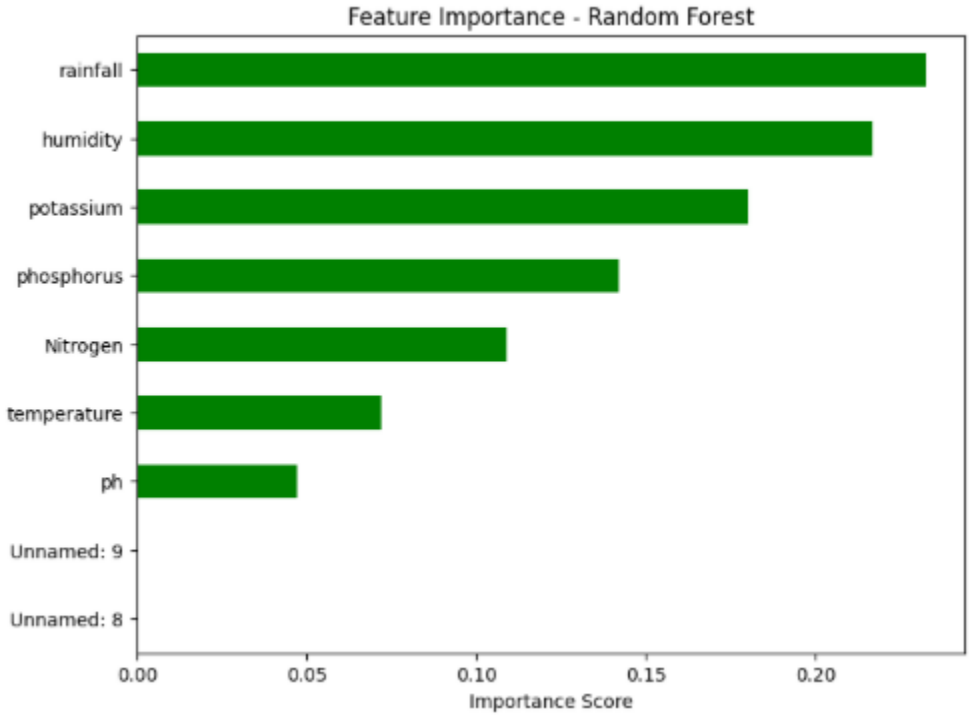
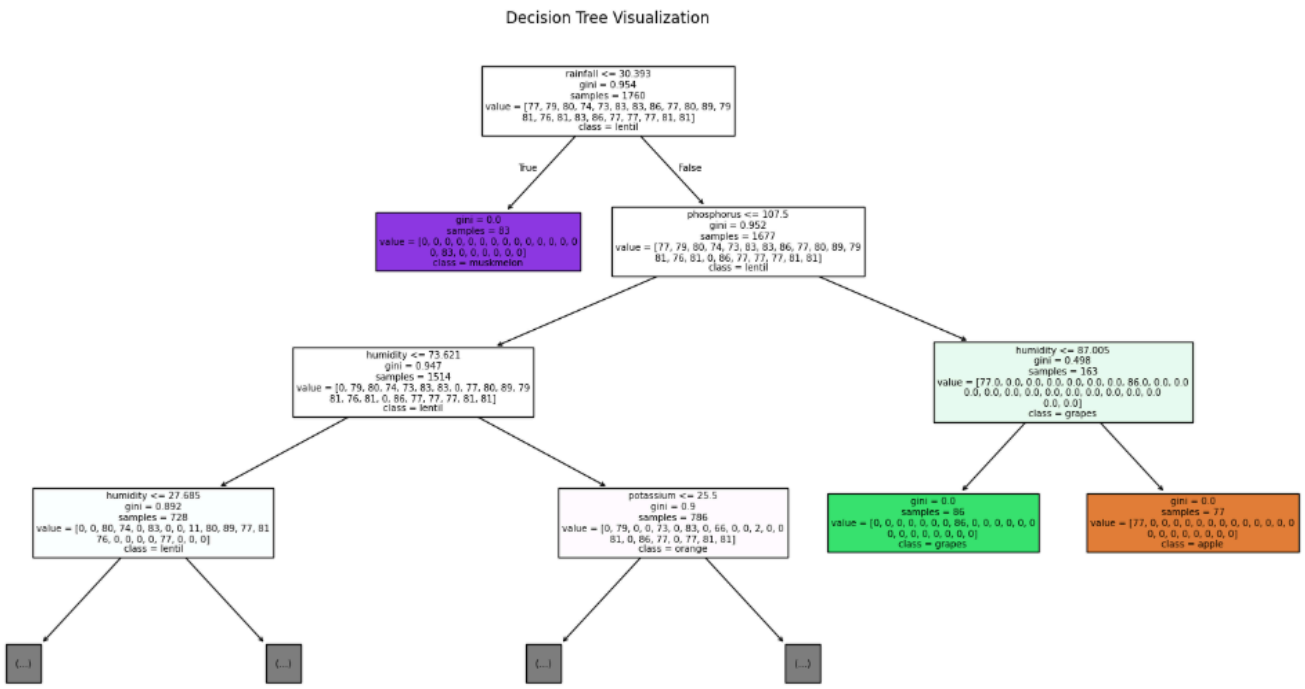
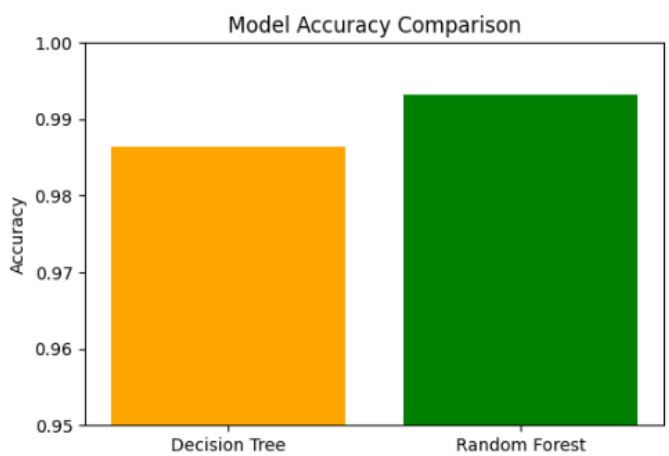


Random Forest Confusion Matrix
Actual
Predicted
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21
0 5 10 15 20 25


Feature Importance - Random Forest
rainfall
humidity
potassium
phosphorus
Nitrogen
temperature
ph
Unnamed: 9
Unnamed: 8
0.00 0.05 0.10 0.15 0.20
Importance Score





Conclusion

Both Decision Tree and Random Forest models performed exceptionally well for crop recommendation, with Random Forest achieving slightly higher accuracy. The most important features influencing predictions were rainfall, temperature, and soil NPK values. Random Forest's ensemble approach reduced overfitting and provided more stable predictions. This experiment demonstrates that machine learning can effectively guide crop selection for better yield and resource management

([collab notebook link](#))